FINITE DIFFERENCE FORMULA FOR THE LEADING COEFFICIENT OF A POLYNOMIAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file intersection.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for this chapter (is-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Lemma 0.1. *Let P be a polynomial of degree r with leading coefficient a. Then*

$$r!a = \sum\nolimits_{i=0,\ldots,r} (-1)^i \binom{r}{i} P(t-i)$$

for any t.

Proof. Let us write Δ the operator which to a polynomial P associates the polynomial $\Delta(P) = P(t) - P(t-1)$. We claim that

$$\Delta^r(P) = \sum\nolimits_{i=0,\ldots,r} (-1)^i \binom{r}{i} P(t-i)$$

This is true for $r = 0, 1$ by inspection. Assume it is true for r. Then we compute

$$\begin{aligned} \Delta^{r+1}(P) &= \sum\nolimits_{i=0,\ldots,r} (-1)^i \binom{r}{i} \Delta(P)(t-i) \\ &= \sum\nolimits_{n=-r,\ldots,0} (-1)^i \binom{r}{i} (P(t-i) - P(t-i-1)) \end{aligned}$$

Thus the claim follows from the equality

$$\binom{r+1}{i} = \binom{r}{i} + \binom{r}{i-1}$$

The lemma follows from the fact that $\Delta(P)$ is of degree $r-1$ with leading coefficient ra if the degree of P is r. $\square$

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